Queries on Operators

1. Find the pname, phoneno and cost of parts which have cost equal to or greater than 200 and less than or equal to 600.

2. Find the sname , SID and branch of suppliers who are in _local‘ branch or _global‘ branch

3. Find the pname, phoneno and cost of parts for which cost is between 200 and 600

4. Find the pname and color of parts, which has the word _NET‘ anywhere in its pname.

5. Find the PID and pname of parts with pname either _NUT‘ or _BOLT‘

6. List the suppliers who supplied parts on _1st may2000‘, _12 JAN 2021‘ ,‘17 dec 2000‘ , ’10 Jan 2021’

7. Find all the distinct costs of parts

1. SELECT p.PName, s.Phone, p.price

FROM Part p, Supplier s, Supplies sp

WHERE p.PID = sp.PID

AND s.SID = sp.SID

AND p.price >= 200

AND p.price <= 600;

PNAME	PHONE	PRICE
BOLT	9611761997	568.68
NUT	9876565677	500

2. SELECT sname, sid, branch

FROM supplier

WHERE branch IN ('Local', 'Global');

SNAME	SID	BRANCH
ARUN	20001	Local
Chandan	2003	Global

Varun	2002	Local
Varsha	2004	Global
Arti	2005	Local

3. SELECT p.PName, s.Phone, p.price

FROM Part p, Supplier s, Supplies sp

WHERE p.PID = sp.PID

AND s.SID = sp.SID

AND p.price between 200

AND 600;

PNAME	PHONE	PRICE
BOLT	9611761997	568.68
NUT	9876565677	500

4. SELECT pname, color

FROM Part

WHERE pname LIKE '%she%';

PNAME	COLOR
washer	black

5. SELECT pid, pname

FROM Part

WHERE pname IN ('NUT', 'BOLT');

PID	PNAME
1001	BOLT
1002	NUT

6. SELECT DISTINCT s.SID, s.SName, s.Phone

FROM Supplier s, Supplies sp where s.SID = sp.SID

and sp.DATE_SUPPLIED IN ('09-AUGUST-2009', '08-june-2020', '09-june-2023');

SID	SNAME	PHONE
2004	Varsha	9855676567
20001	ARUN	9611761997
2003	Chandan	9876565677

7. SELECT DISTINCT price

FROM Part;

PRICE
568.68
500
700
699
800

Queries on Set Theory Operators

1. List all parts except _NUT‘ and _BOLT‘ in ascending order of costs

2. display all parts that have not been supplied so far

3. To display the supplier names who have supplied _green‘ part with cost 500 Rupees AND _red‘ part with cost 400 Rupees.

4. To display the supplier names who have supplied _green‘ part with cost 500 Rupees OR _red‘ part with cost 400 Rupees.

5. To Display the name of suppliers who have supplied all parts that are _red‘ in color.

1. SELECT pid, pname, price

FROM part

WHERE pname NOT IN ('NUT', 'BOLT')

ORDER BY price ASC;

PID	PNAME	PRICE
1003	washer	699
1004	screw	700
1005	plate	800

2. SELECT pid, pname, price

FROM part

MINUS

SELECT p.pid, p.pname, p.price

FROM part p

JOIN supplies sp ON p.pid = sp.pid;

(Uses MINUS → finds parts not appearing in supplies.

delete from supplies where PID=1005;)

PID	PNAME	PRICE
1005	plate	800

3. SELECT s.sname

FROM supplier s

JOIN supplies sp ON s.sid = sp.sid

JOIN part p ON sp.pid = p.pid

WHERE p.color = 'silver' AND p.price = 500

INTERSECT

SELECT s.sname

FROM supplier s

JOIN supplies sp ON s.sid = sp.sid

JOIN part p ON sp.pid = p.pid

WHERE p.color = 'red' AND p.price = 700;

no data found (insert employee who supplies both color then check)

4. SELECT s.sname

FROM supplier s

JOIN supplies sp ON s.sid = sp.sid

JOIN part p ON sp.pid = p.pid

WHERE p.color = 'silver' AND p.price = 500

union

SELECT s.sname

FROM supplier s

JOIN supplies sp ON s.sid = sp.sid

JOIN part p ON sp.pid = p.pid

WHERE p.color = 'red' AND p.price = 700;

SNAME
Chandan
Varsha

5. SELECT s.sname

```
FROM supplier s
WHERE NOT EXISTS (
   SELECT p.pid
   FROM part p
   WHERE p.color = 'red'
   MINUS
   SELECT sp.pid
   FROM supplies sp
   WHERE sp.sid = s.sid
);
```

SNAME
Varsha

PLSQL

1. Write a PL/SQL Code to add two numbers

```
DECLARE
a NUMBER := 10;
b NUMBER := 20;
total NUMBER;
BEGIN
total := a + b;
DBMS_OUTPUT.PUT_LINE('Sum of two numbers = ' || total);
END;
```

Sum of two numbers = 30

Statement processed.

2. Write a PL/SQL code for Fibonacci series

```
DECLARE
n NUMBER := 10; -- number of terms
a NUMBER := 0;
b NUMBER := 1;
c NUMBER;
BEGIN
DBMS_OUTPUT.PUT_LINE('Fibonacci Series:');
DBMS_OUTPUT.PUT_LINE(a);
DBMS_OUTPUT.PUT_LINE(b);
FOR i IN 3..n LOOP
c := a + b;
DBMS_OUTPUT.PUT_LINE(c);
a := b;
b := c;
END LOOP;
END;
```

```
Fibonacci Series:
0
1
1
2
3
5
8
13
21
34

Statement processed
```

3. Write a PL/SQL Code for greatest of 3 numbers

```
DECLARE
```

```
a NUMBER := 25;
b NUMBER := 40;
c NUMBER := 30;
greatest NUMBER;
BEGIN
IF a > b AND a > c THEN
greatest := a;
ELSIF b > c THEN
greatest := b;
ELSE
greatest := c;
END IF;
DBMS_OUTPUT.PUT_LINE ('Greatest Number = ' || greatest);
END;
```

```
Greatest Number = 40
Statement processed.
```

4. Write a PL/SQL code for area and circumference of a circle

```
DECLARE
radius NUMBER := 5;
area NUMBER;
circumference NUMBER;
pi CONSTANT NUMBER := 3.14159;
BEGIN
area := pi * radius * radius;
circumference := 2 * pi * radius;
DBMS_OUTPUT.PUT_LINE ('Radius = ' || radius);
```

DBMS_OUTPUT.PUT_LINE('Area of Circle = ' || area);

DBMS_OUTPUT.PUT_LINE('Circumference of Circle = ' || circumference);

END;

```
Radius = 5
Area of Circle = 78.53975
Circumference of Circle = 31.4159

Statement processed
```